

Turtle Games

Customer Loyalty & Sentiment Analysis



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1. Background

Problem Statement

Turtle Games' (TG) sales have stagnated. Likely causes include marketing processes that fail to leverage the loyalty program. Customers accumulate points but the associated data is not used in targeted initiatives. Additionally, customer reviews have not been evaluated for sentiment. These factors combined leave TG with underdeveloped insights and marketing strategies.

Objective

Build models that predict loyalty and gauge sentiment from reviews to inform new marketing strategies, increase engagement and boost sales.

Appendix A: 'Five Whys' analysis of business problem

2. Analytical Approach

2.1 Data Preparation and Exploratory Analysis (EDA)

Completed in R due to historic use in TG.

- Imported “turtle_reviews.csv”, examined column names, structure
- Integrity check - data types, nulls, missing values etc
- Separated ‘review’ and ‘summary’ for sentiment analysis, keep remaining data for EDA and modelling
- Punctuation removed from column names, simple naming convention applied
- Converted ‘education’ into ordinal categories
- Used ggplot2 for exploratory visualisations;
 - Histogram - loyalty points distribution
 - Scatter plots - spending, income and loyalty relationships
 - Boxplots – loyalty in demographic groupings
- Exported data sets for analysis in Python
 - “turtle_ready_for_regression.csv”
 - “turtle_review_text.csv”

2.2 Initial Predictive Modelling: All Customers (Regression)

Leverage Python ML libraries

- Import “turtle_ready_for_regression.csv” as data frame
- Defined ‘loyalty’ as dependent variable (y)
- Repeated linear regression exercise with ‘x’ set to ‘spending’, ‘income’ then ‘age’
- Evaluated models with R-squared
- Visualised using scatter plots with regression line

2.3 Clustering

- Selected ‘spending’ and ‘income’ as clustering features (strongest available predictors of loyalty)
- Pair plots and scatterplots to observe groupings
- Elbow and silhouette methods confirm group number
- K-means identified distinct segments
- Labelled clusters 0 – 4 in new column. Created separate data frames

2.4 Targeted Predictive Modelling

Higher value segments analysed;

- Cluster_0
- Cluster_1
- Cluster_3

Decision trees fitted to Cluster_0 and Cluster_3. Cluster_1 omitted.

- Features: income, spending (X), loyalty (y)
- Shallow tree (max_depth=3) chosen due to;
 - Small size of dataset
 - Only two features selected
 - Avoid risk of overfitting
- Create test/train sets, Decision Tree Regressor, make predictions and evaluate (MSE, R^2)
- View tree as text and visualisation

2.5 Customer Reviews: Natural Language Processing

Create Wordclouds from 'review' and 'summary'

- Tokenise and flatten columns into text strings
- Observe frequency of tokens
- Remove non-alphabetical characters and stopwords
- Generate wordclouds

Opted for VADER compound scores, better optimised for informal, nuanced text

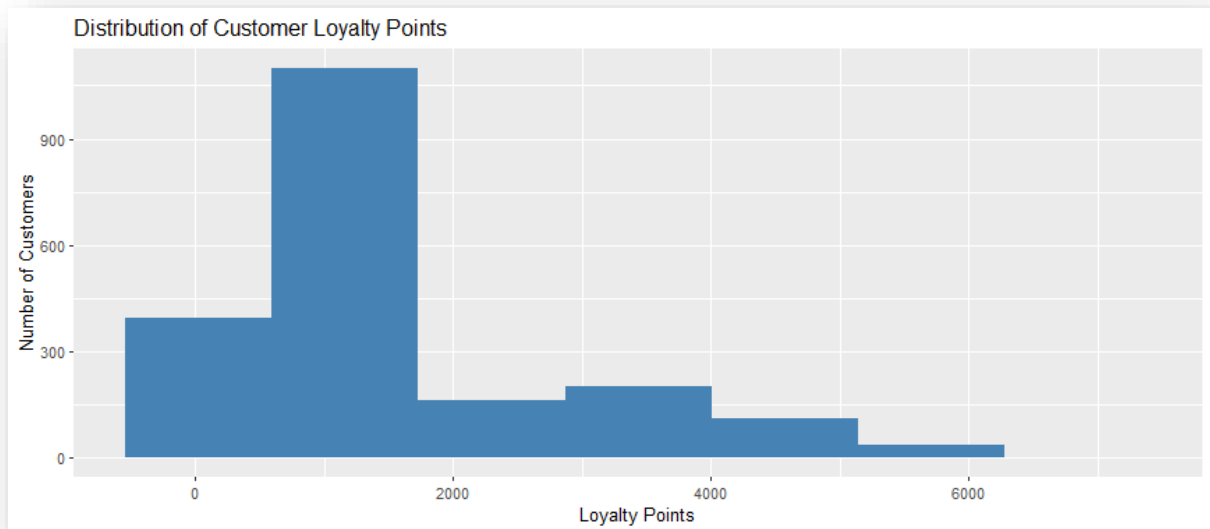
- Import relevant packages
- Calculate compound scores
- Generate histograms showing score distribution
- Print most negative and most positive reviews/summaries

3. Visualisations and Insights

3.1. Exploratory Analysis

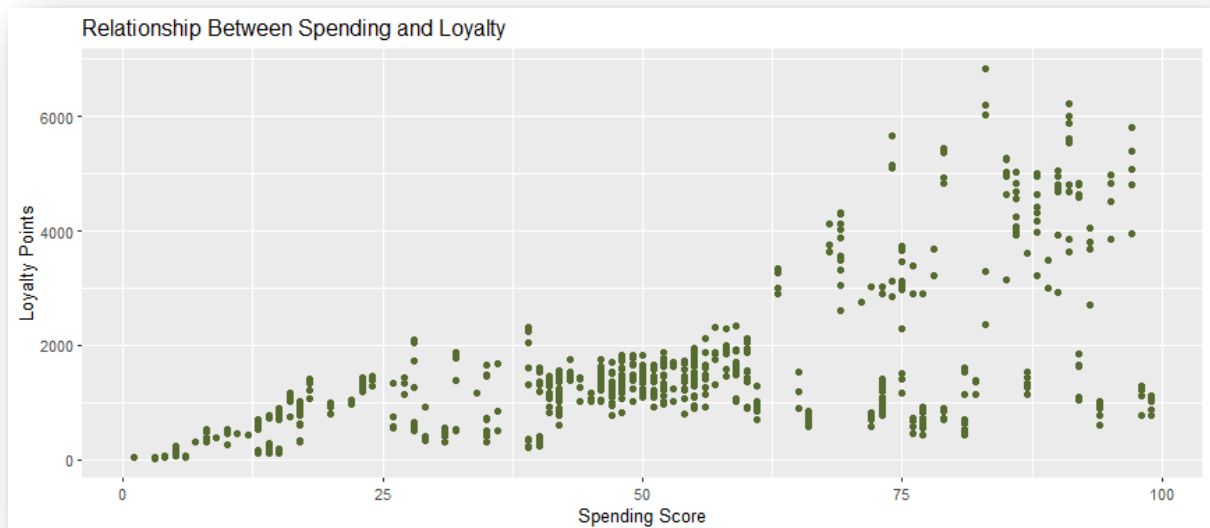
Visualise surface level insights (R).

Figure 1: Distribution of Loyalty Points



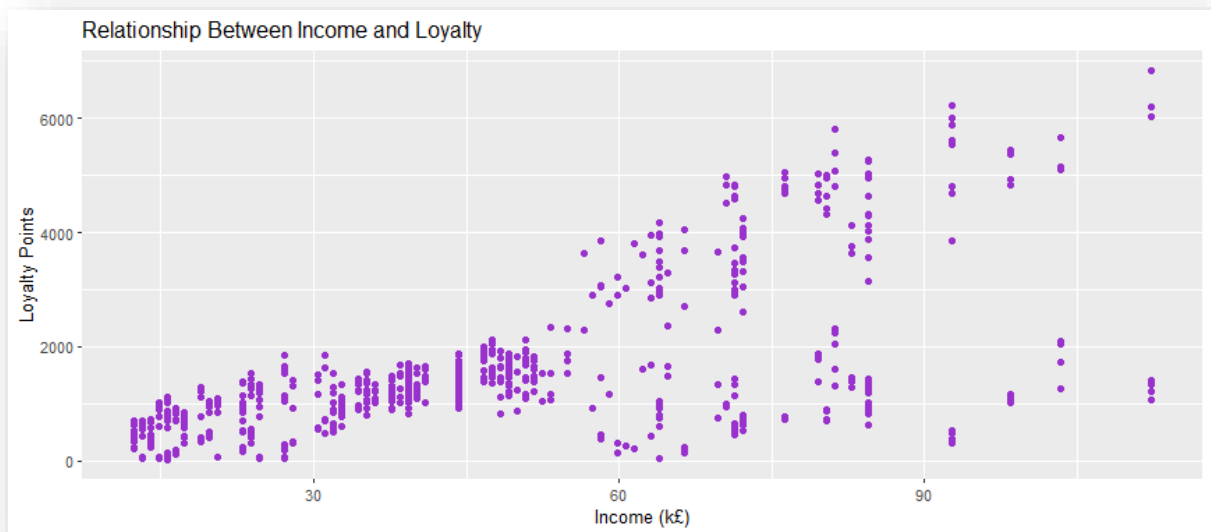
Loyalty distribution is right-skewed. Most customers accumulate < 2,000 points.

Figure 2: Spending vs Loyalty



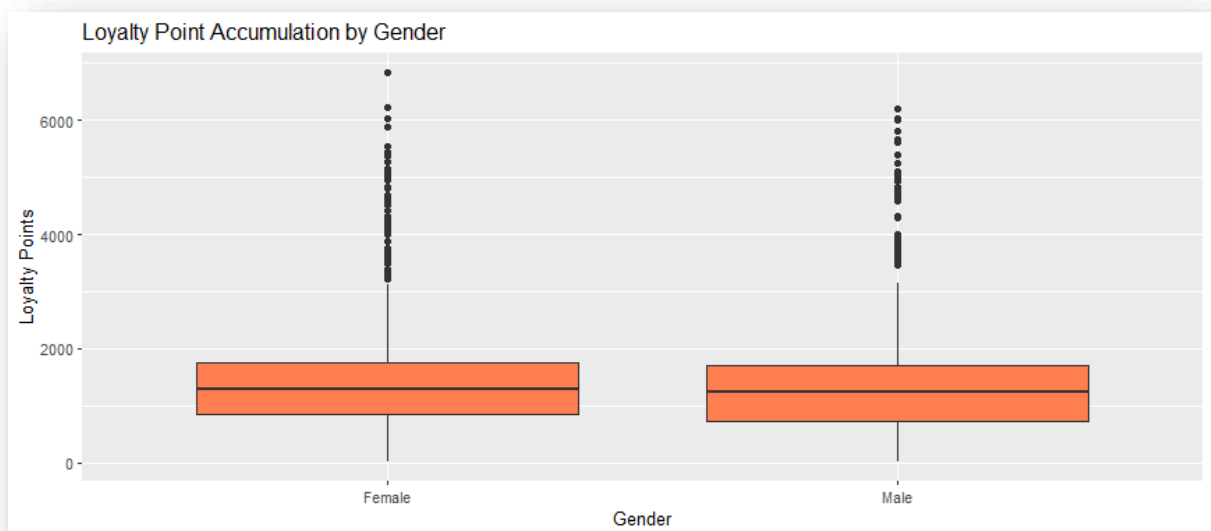
Spending shows a positive trend. Generally, higher spending corresponds with higher loyalty point accumulation.

Figure 3: Income vs Loyalty



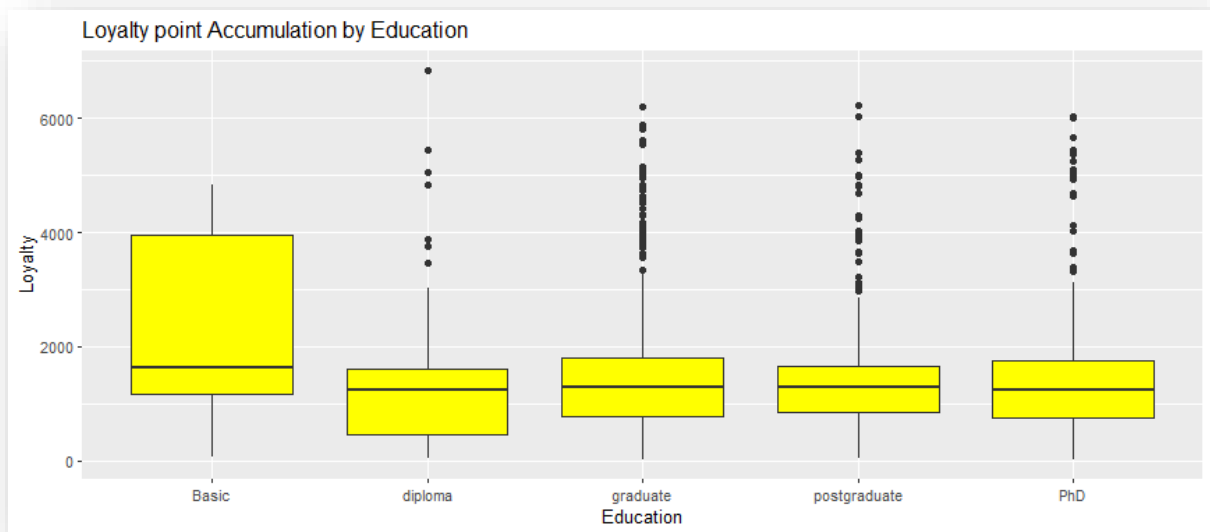
Some correlation between income and loyalty, not as strong as spending.

Figure 4: Loyalty by Gender



Gender does not significantly influence loyalty.

Figure 5: Loyalty by Education

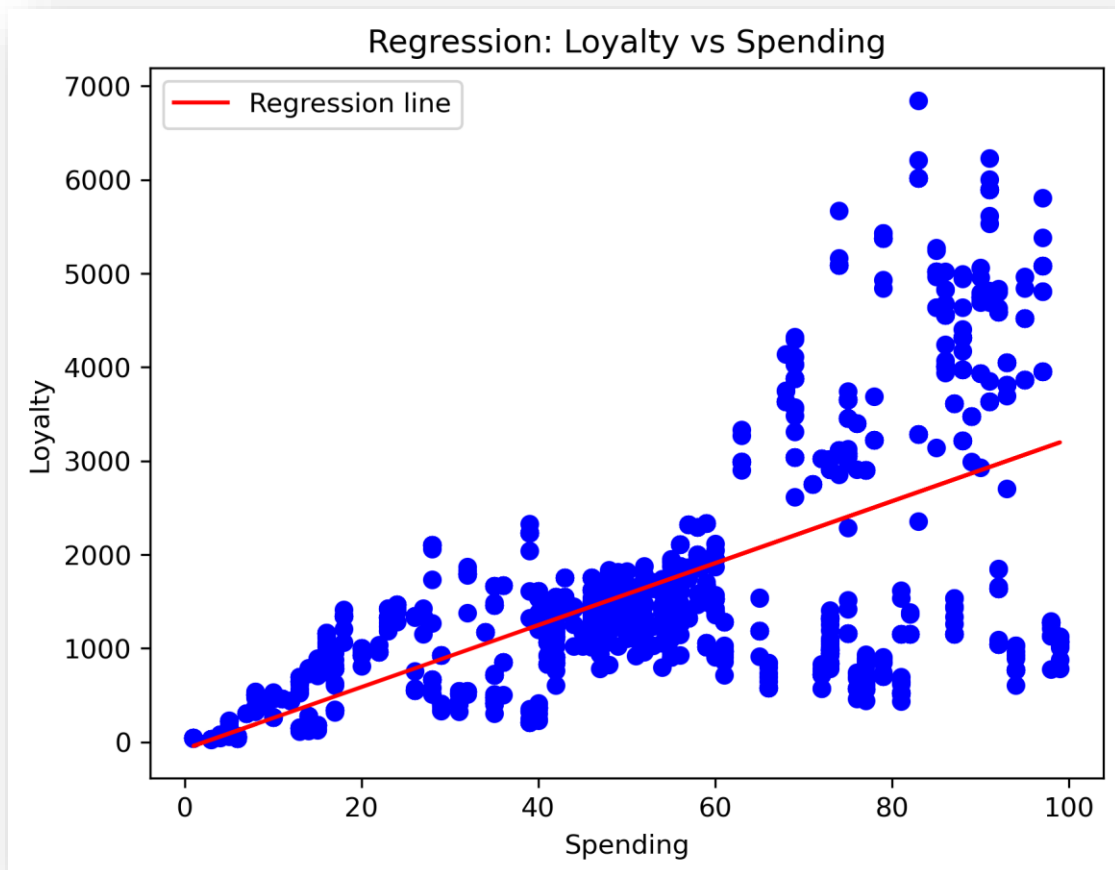


Highest median loyalty in “Basic”. Other levels show lower medians with outliers indicating increasingly loyal sub-groups.

3.2 Initial Predictive Modelling: All Customers

Using Simple Linear Regression, assess whether loyalty can be predicted from spending, income or age.

Figure 6: Spending vs Loyalty

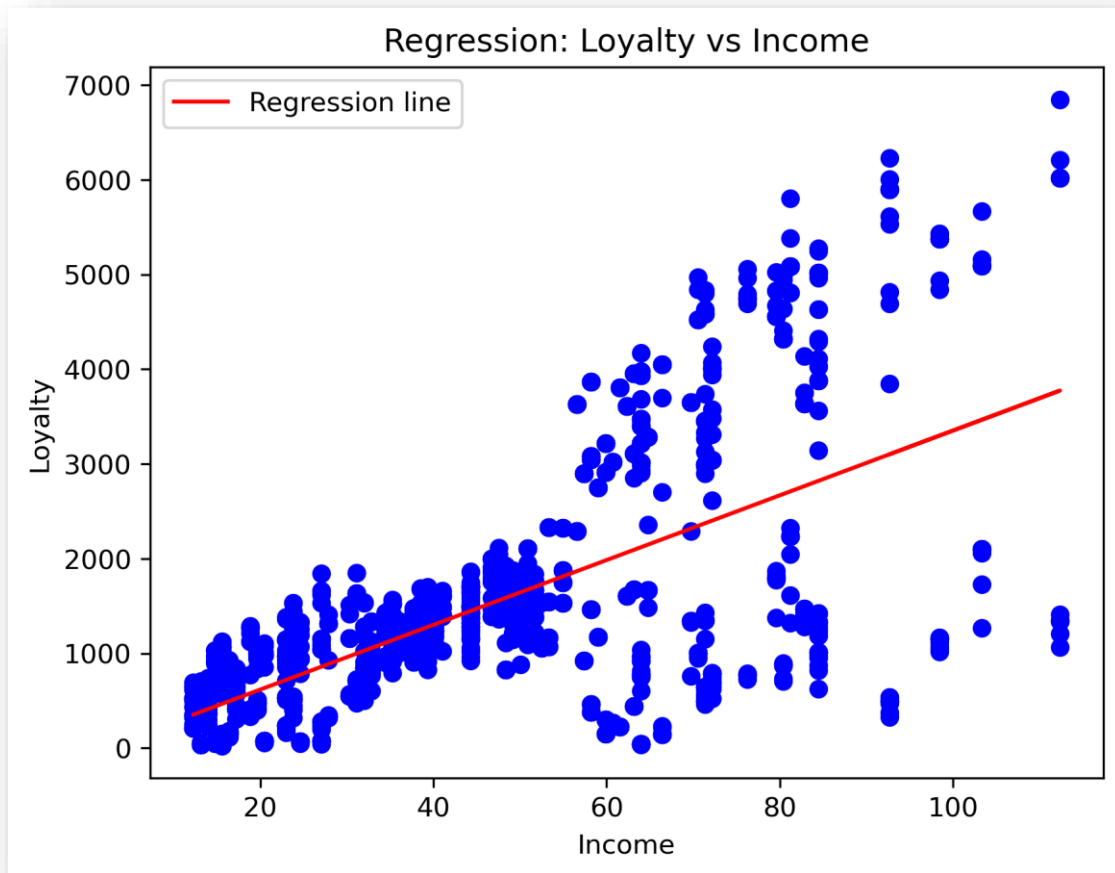


Moderate positive linear relationship ($R^2 = 0.45$).

£1 spent generates approx. 33 points

Heteroscedasticity observed as spending increases.

Figure 7: Income vs Loyalty

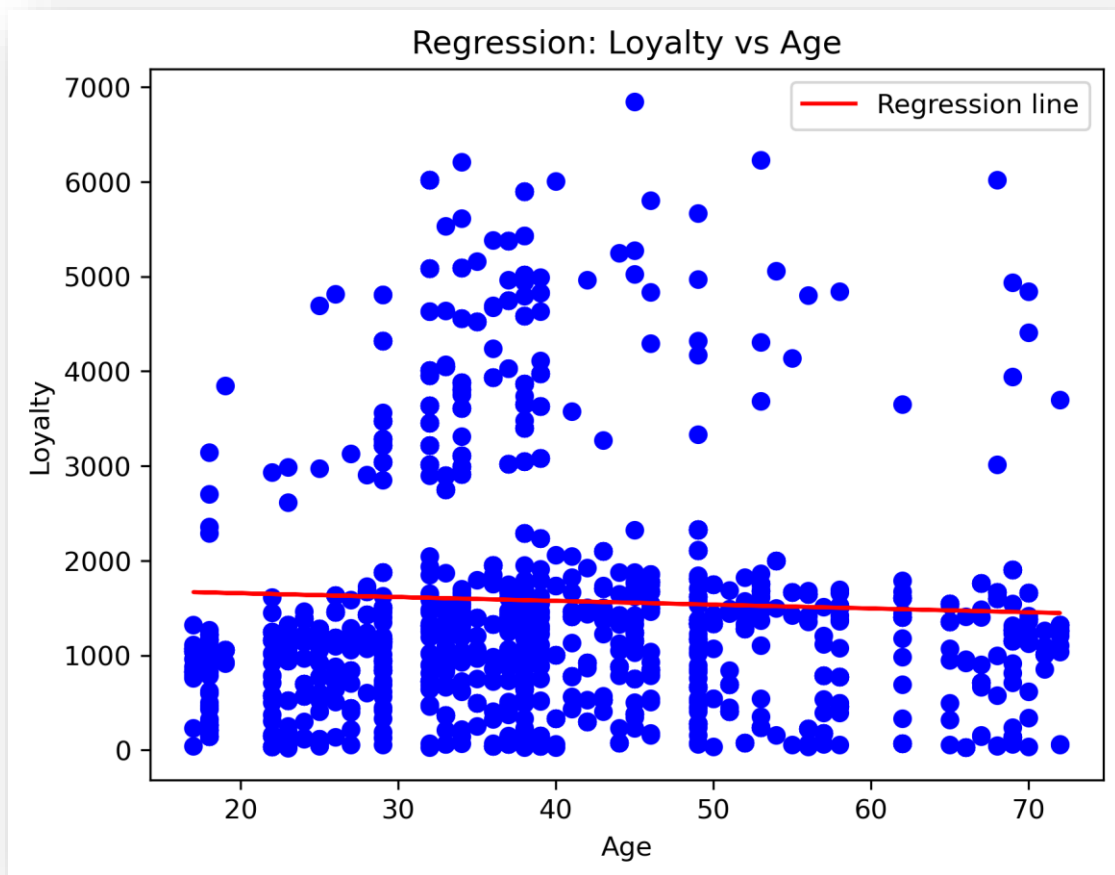


Weaker moderate positive relationship between income and loyalty ($R^2 = 0.38$).

£1000 salary increase yields approx. 34 points

Heteroscedasticity present.

Figure 8: Age vs Loyalty



Age not a meaningful predictor ($R^2 = 0.002$) and dropped.

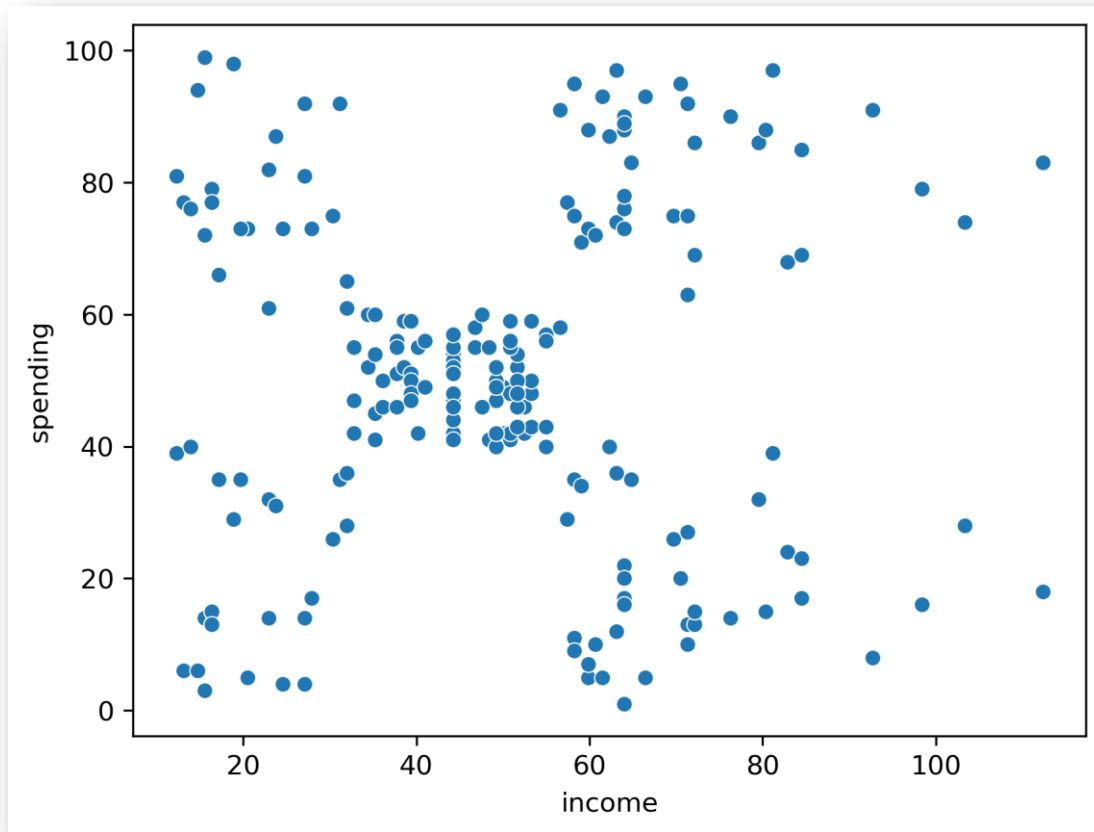
Spending and income show moderate relationship with loyalty. Decision to identify customer segments, enabling stronger predictions.

Appendix B: Regression statistics

3.3 Clustering

Visualisations to track identification, validation, model fit and labelling processes.

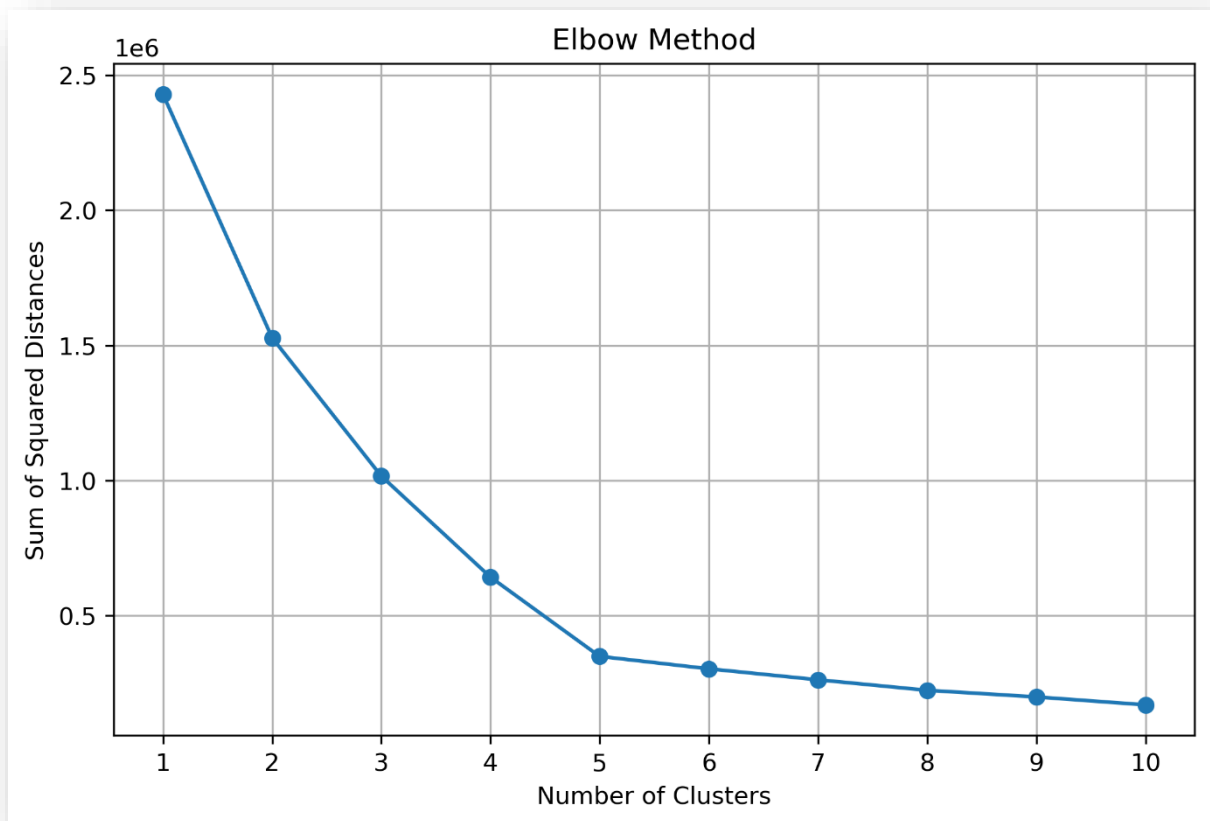
Figure 9: Income vs Spending



Observed five homogenous groups

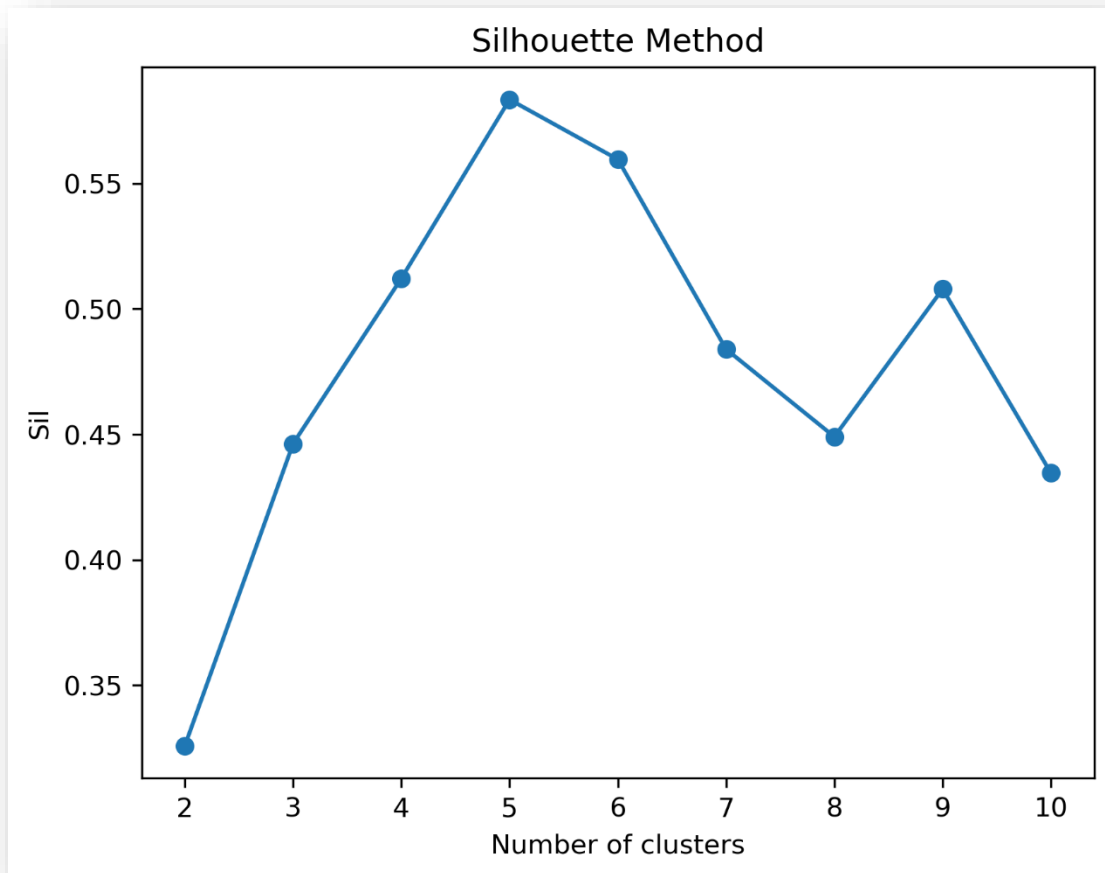
Appendix C: More clustering data

Figure 10: Identify number of clusters



Elbow at $k = 5$ indicating five clusters

Figure 11: Validate number of clusters



Number confirmed using silhouette method

Figure 12: Customer Segments

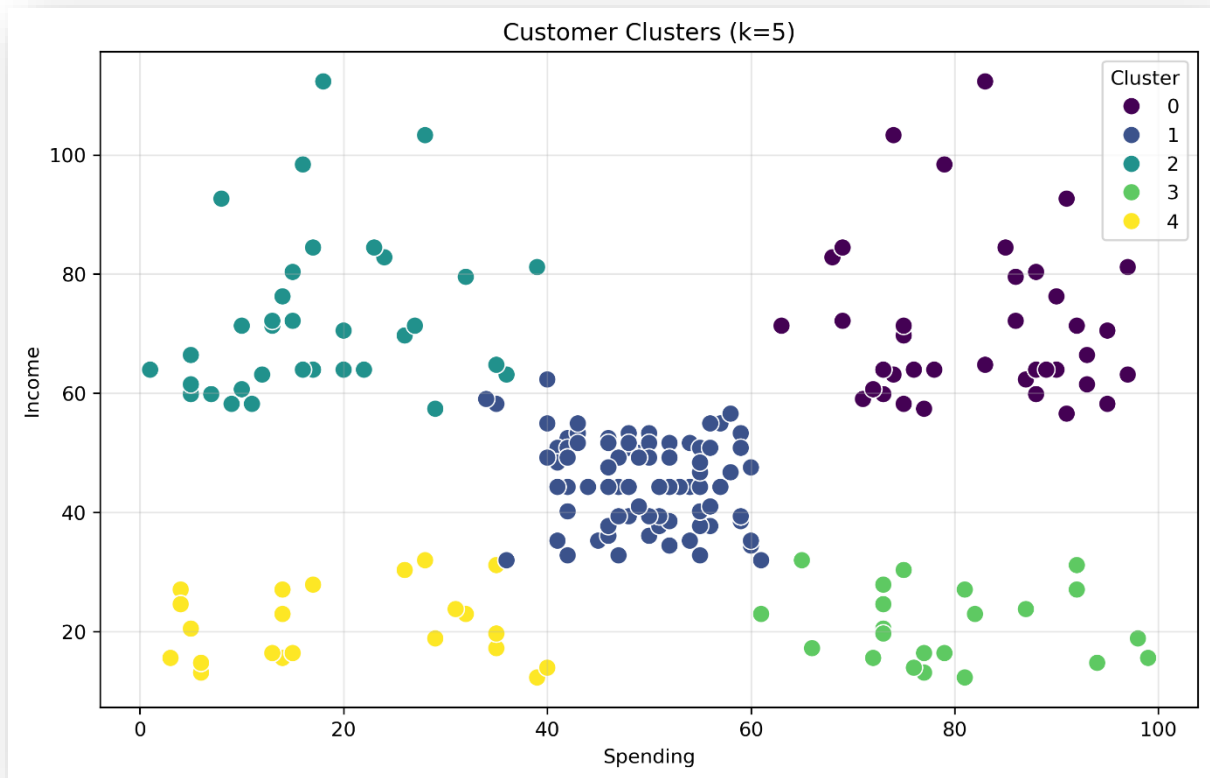


Table 1: Segment Descriptions

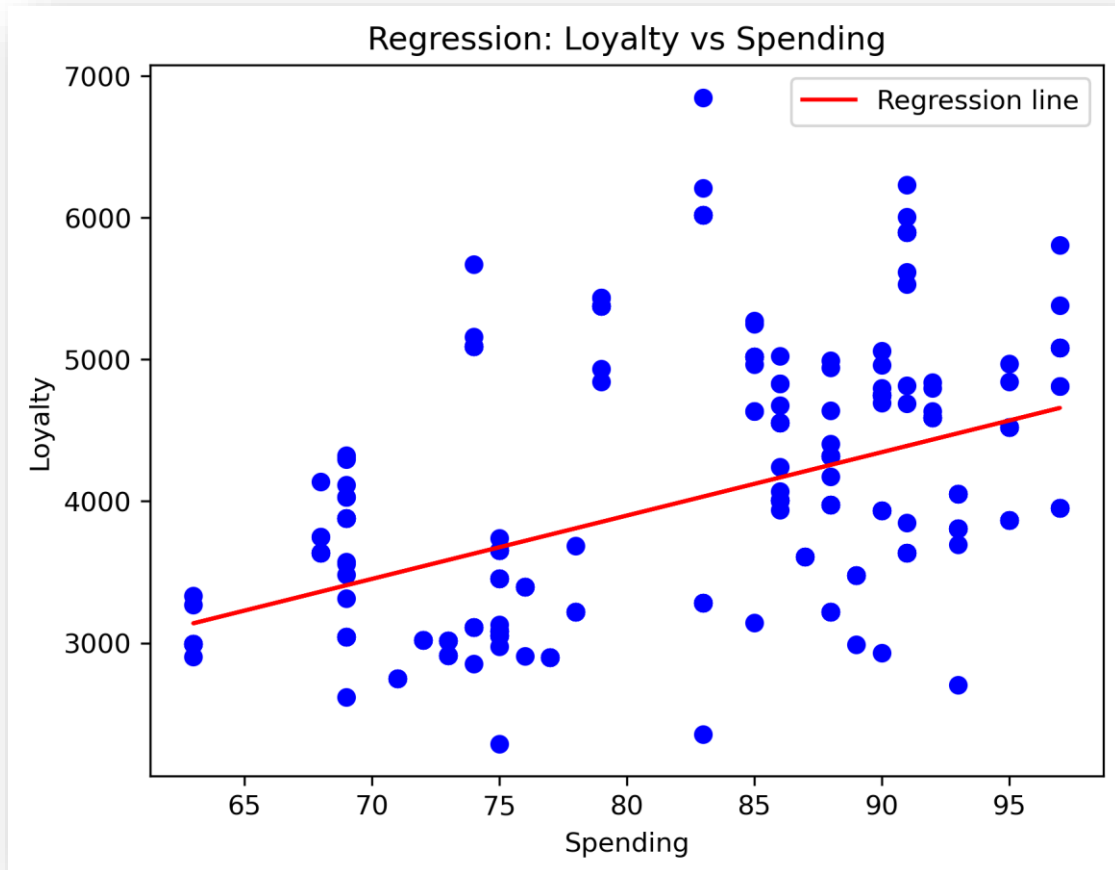
Label	Description	Persona	Note
Cluster_0	High income, high spend	Executive Collectors	High value segment
Cluster_1	Middle income, mid spend	Busy Shoppers	Low value segment
Cluster_2	High income, low spend	Selective Spenders	Low value segment
Cluster_3	Low income, high spend	Generous Parents	High value segment
Cluster_4	Low income, low spend	Budgeting Parents	Low value segment

Appendix D: Segmentation descriptions & assumptions

3.4 Targeted Predictive Modelling

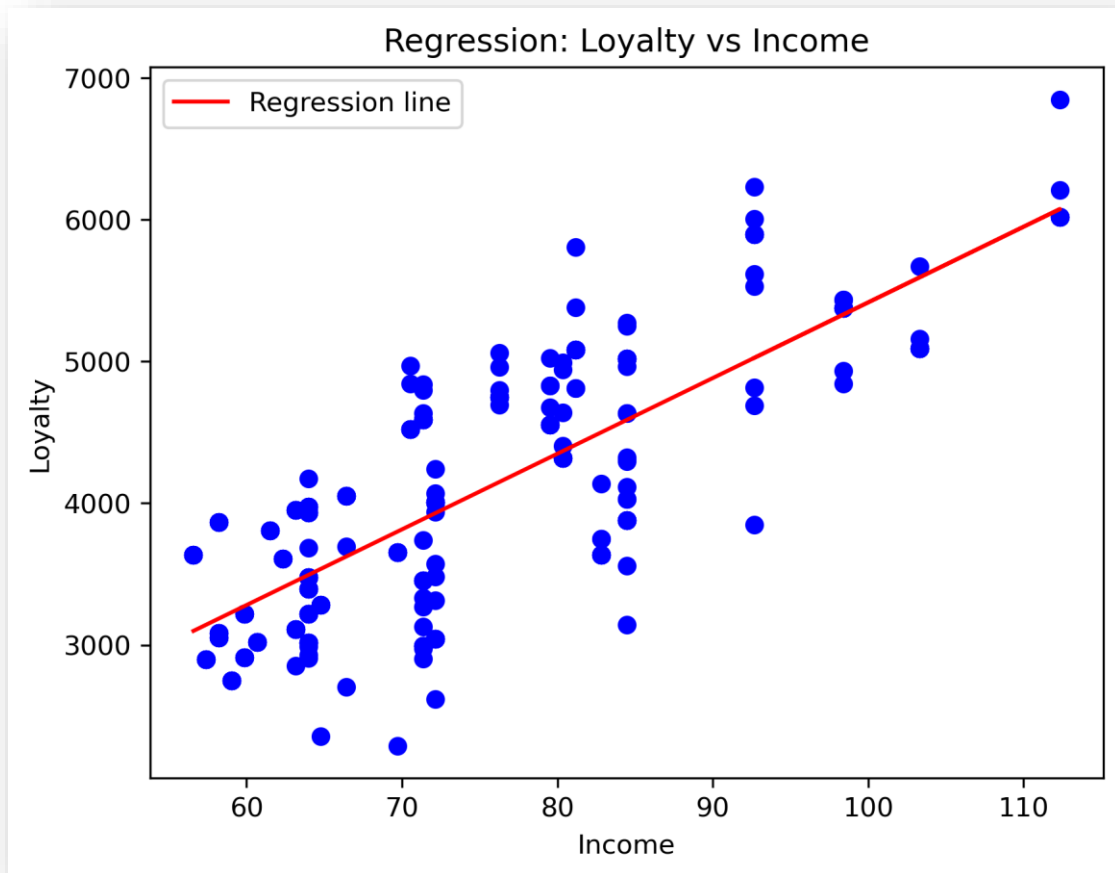
3.4a Simple Linear Regression repeated on clusters 0, 1, 3

Figure 13: Cluster 0 Spending vs Loyalty



Spending weak predictor of loyalty ($R^2 = 0.216$).

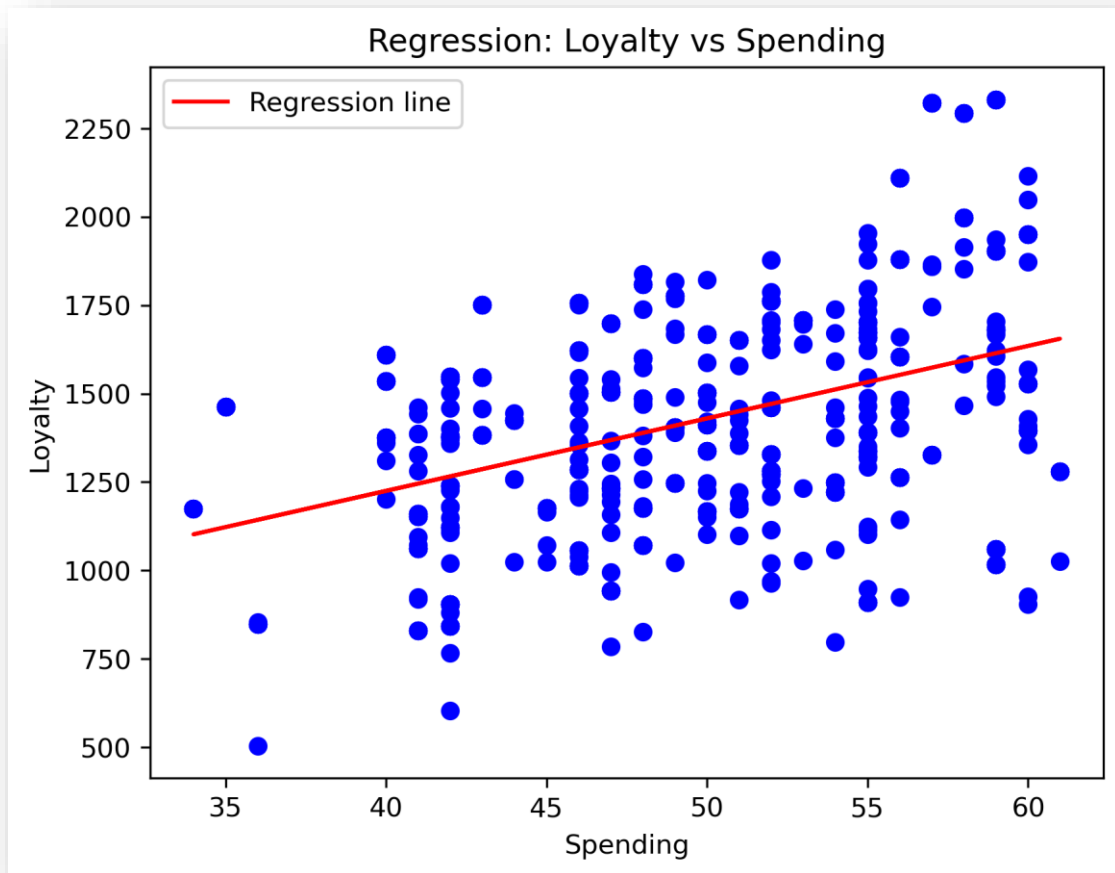
Figure 14: Cluster 0 Income vs Loyalty



Income stronger predictor of loyalty for this group ($R^2 = 0.649$).

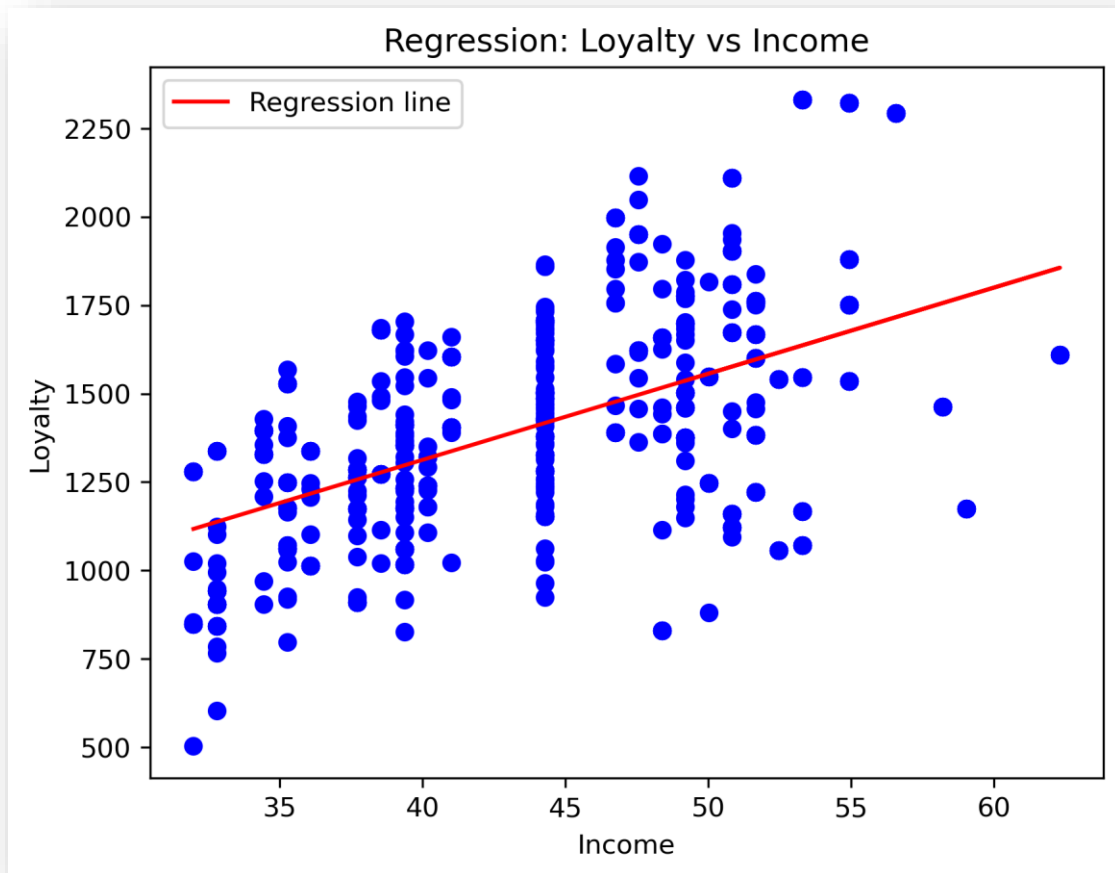
£1000 salary increase yields approx. 53 points

Figure 15: Cluster 1 Spending vs Loyalty



Spending very weak predictor ($R^2 = 0.17$).

Figure 16: Cluster 1 Income vs Loyalty



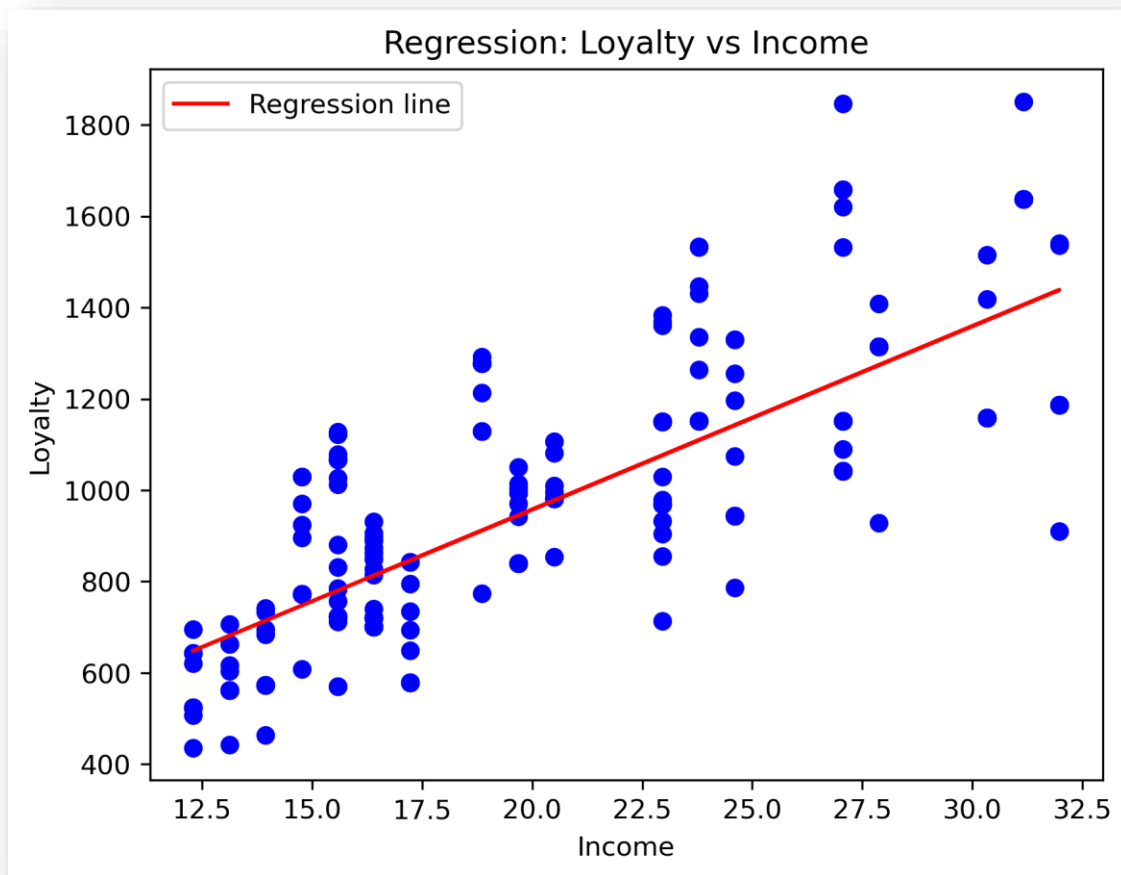
Income insufficiently strong predictor ($R^2 = 0.286$). Drop cluster_1 for next stage.

Figure 17: Cluster 3 Spending vs Loyalty



Spending very weak predictor of loyalty for this group ($R^2 = 0.089$).

Figure 18: Cluster 3 Income vs Loyalty

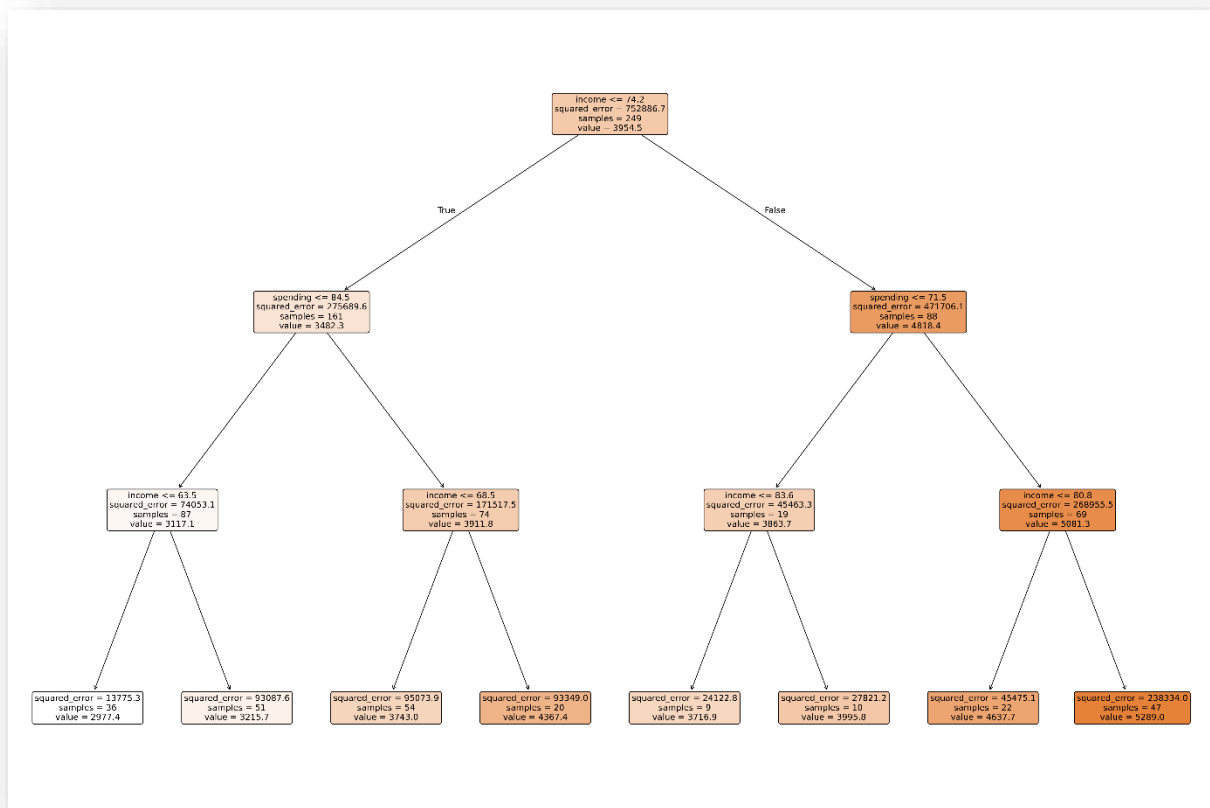


Income stronger predictor of loyalty for this group ($R^2 = 0.605$).

£1,000 increase yields approximately 40 points

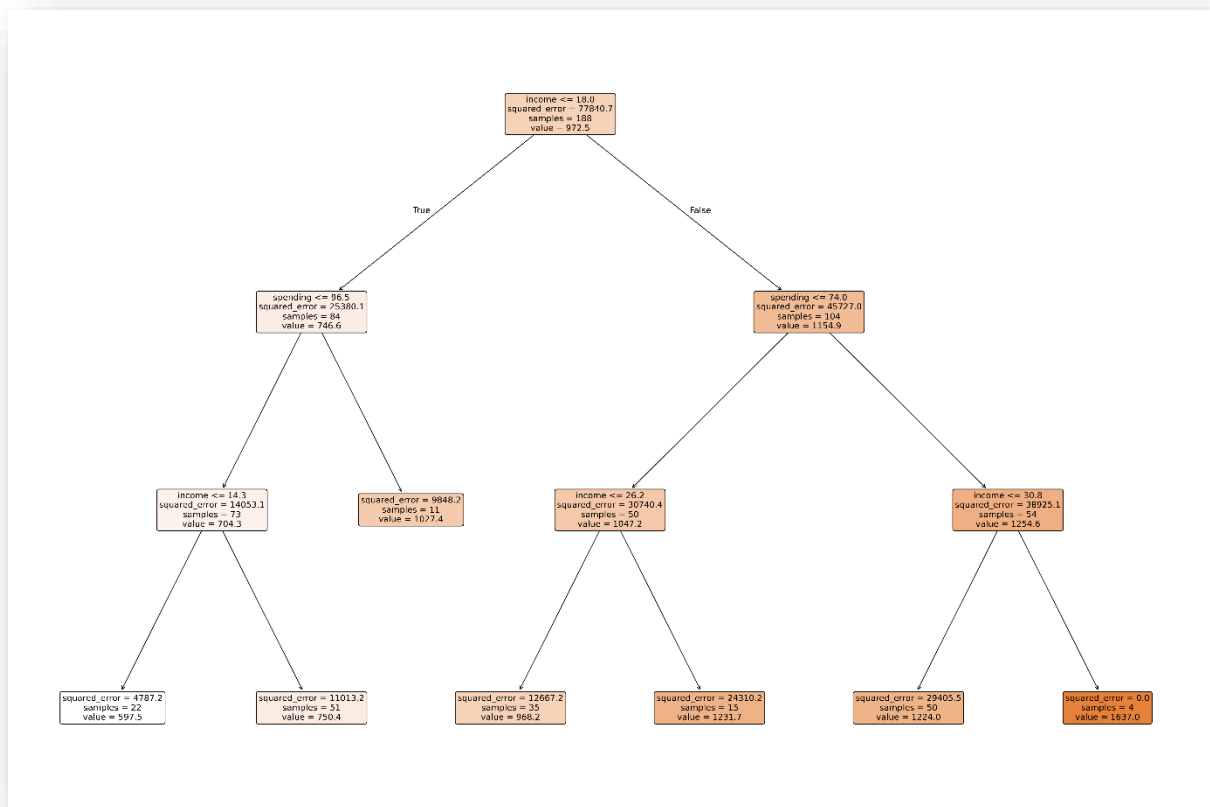
3.4b Decision trees (features 'income' and 'spending'), clusters 0 and 3

Figure 19: Cluster 0



Drivers for loyalty thresholds can be accurately predicted for this group ($R^2 = 0.83$)

Figure 20: Cluster 3



Equal accuracy achieved for this group ($R^2 = 0.83$)

Outcome: accurate models predict loyalty accumulation thresholds for the highest value groupings.

Appendix E: Trees printed in text format

3.5 Customer Reviews: Natural Language Processing

Figure 21: Reviews Wordcloud

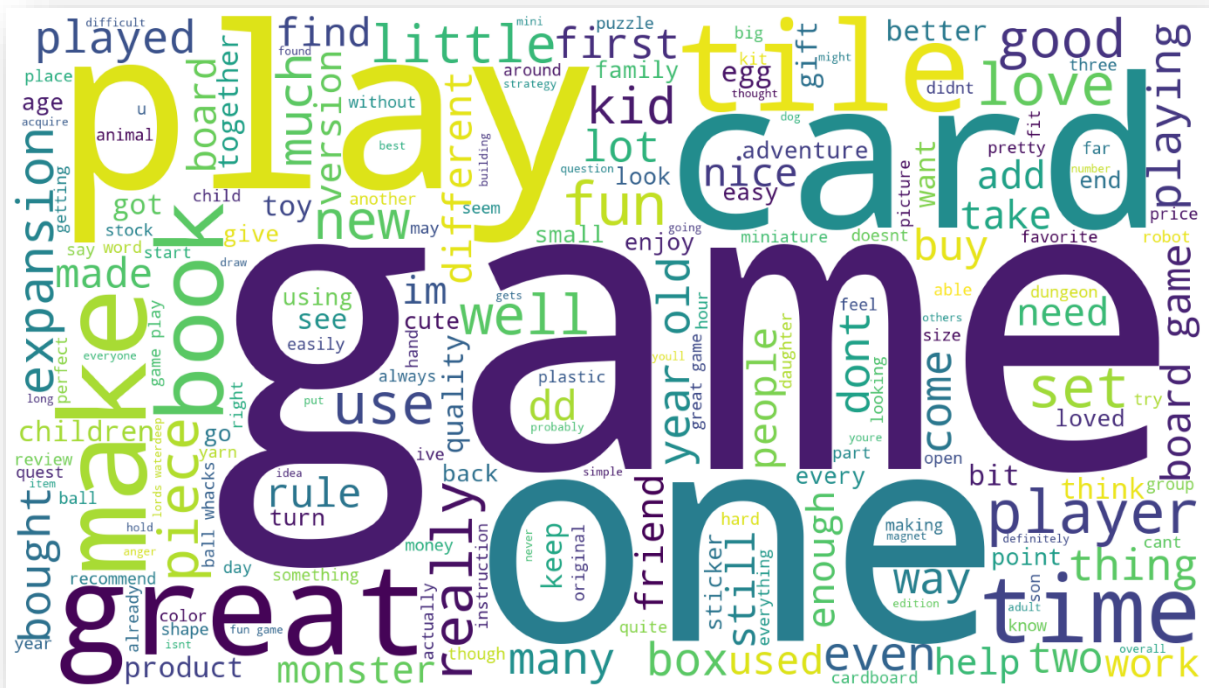
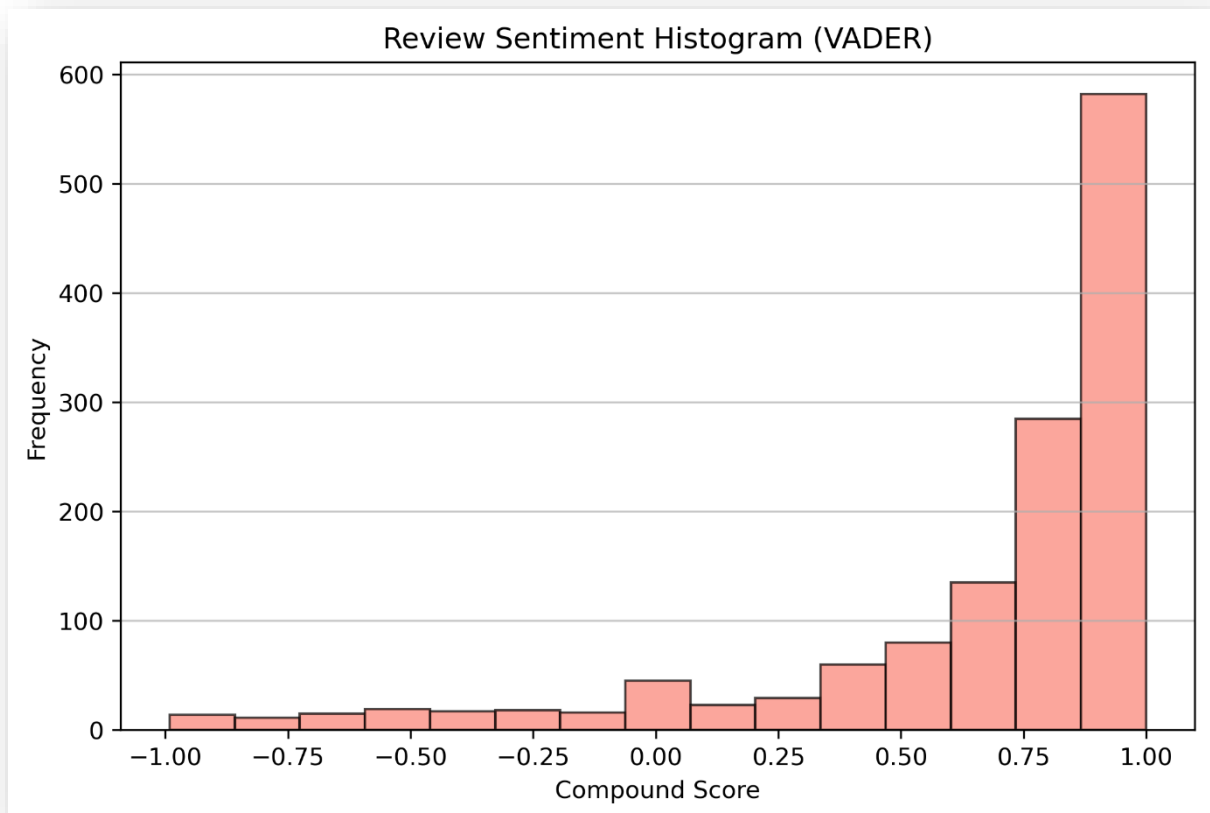


Figure 22: Summaries Wordcloud



Wordclouds impactful but lack detailed insights. More detailed analysis of customer sentiments is required to understand feedback at scale.

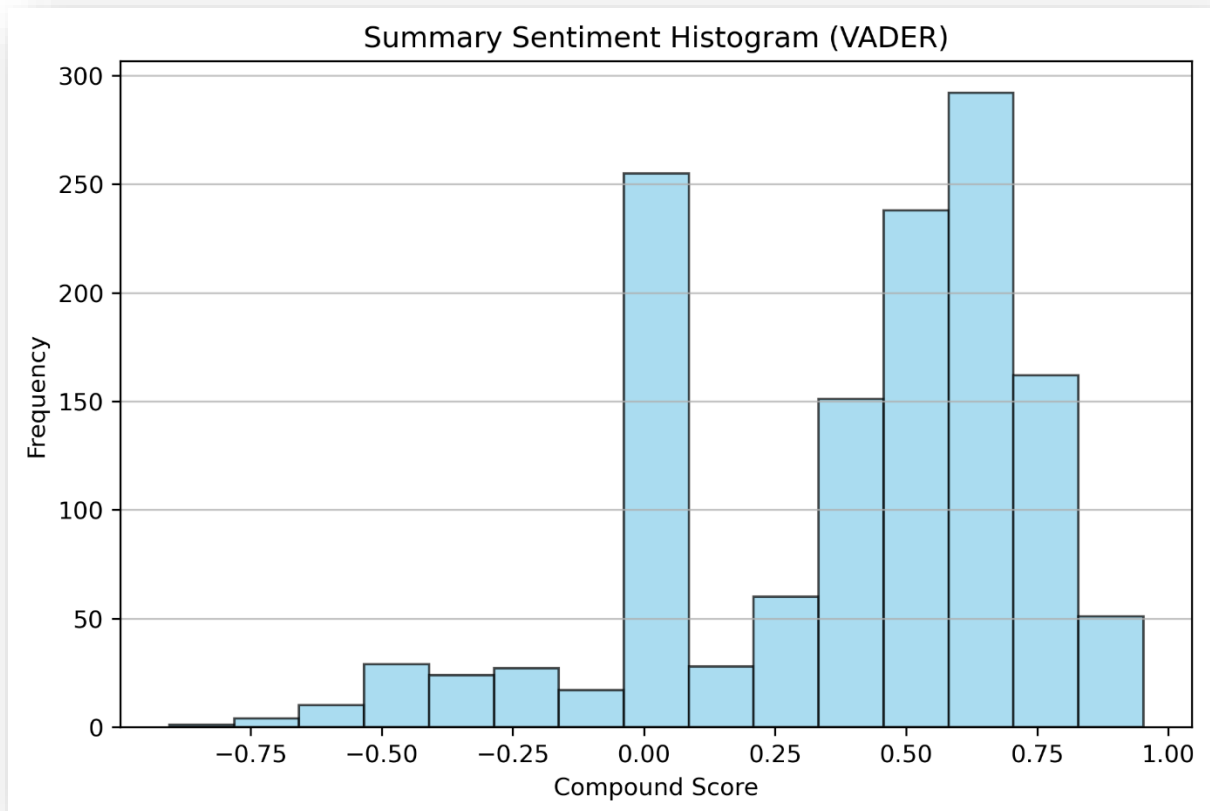
Figure 23: Vader compound sentiment distribution – reviews



Vader compound score provide greater statistical analysis of text.

Reviews overwhelmingly positive. Most scores => 0.5.

Figure 23: Vader compound sentiment distribution – reviews



Summaries positive overall. These are “review titles” and likely to contain more neutral language.

4. Interpretations, Predictions & Recommendations

Interpretations - Translating observed patterns and insights

Exploratory Analysis;

- Most customers have fewer than 2000 loyalty points, which can be interpreted as a clear signal to boost engagement
- Because spending then income were the strongest predictors of loyalty found in the data, they were selected as features for predictive analysis

Full data set regression;

- Confirmation of results from EDA process: spending and income are strongest drivers of loyalty whereas age does not provide basis for predictive analysis
- That said, their explanatory power remained moderate (<50%). This observation drove the decision to move to a more granular, segmented approach

Clustering outcomes;

- Five distinctly homogenous customer segments were identified enabling their development into descriptive marketing personas
- Of those, three higher-value groups were advanced for targeted modelling

Targeted predictive modelling;

- Regression on high value segments revealed income as the strongest predictor, in contrast from the "all-customer" exercise (spending), suggesting contrasting motivations for these groups
- Decision trees confirm the "motivation" hypothesis with clear thresholds that appear to drive increased loyalty

NLP results;

- Overwhelmingly positive sentiment indicates a receptive customer base and lowers risk for new marketing initiatives
- Negative reviews are actionable and provide opportunities for improvements

Predictions

Early models predicted a "general rule" of loyalty rising with spending, but were insufficient for more detailed predictive analysis.

High precision modelling with distinct groups reveal;

- Cluster 0: loyalty highest when income > £80,770 and spending > £71.50
- Cluster 3: spending > £74.00 drives loyalty, boosted further by income > £30,750.

Recommendations

Cluster_0:

- Launch “TG Gold Collectors” inviting customers meeting spending/income thresholds exclusively
- Membership includes;
 - Early access to high-end products
 - Exclusive events, eg RPG evenings hosted at historic venues
 - Exclusive online “unboxing” videos
 - Collector’s Edition products
 - Personalised contact from “Specialist Curator”
 - Limited edition collectible every 3000 points

Cluster_3:

- Launch “Turtle Value Club”
- Discounted bundles to encourage spending > £74
- Loyalty point multipliers on purchases > £74
- Exclusive offers targeted at customers earning > £30750
- Harness pester power advertising during content parents and children watch together
- Double points after three consecutive transactions => £74
- Discounted cinemas tickets every 1000 points
- Free gift every 3000 points

Clusters 1, 2 & 4

- Introduce “TG Celebrations” gift reminder scheme;
 - Online calendar to save gift dates
 - Bonus points for purchases via reminder notification
 - Free greeting card every 1000 points

5. Appendix

Appendix A: 'Five Whys' analysis of business problem

Why does Turtle Games need to boost sales?

Assume that sales have stagnated

Why have sales stagnated?

Assume that TG has failed to understand:

- How to leverage its loyalty points system optimally
- The potential for segmenting customers into groups

Why address this now?

Assume TG has matured to the point where it is ready to evaluate marketing data

Why is it ready now?

The data set appears to contain features that can be used for predictive data analysis

Why do you think the data is suitable?

We have sufficient demographic, loyalty and sentiment data to be able to identify customer insights and make recommendations that answer the following business questions;

- How do customers engage with and accumulate loyalty points?
- How can customers be segmented into groups, and which groups can be targeted by the marketing department?
- How can text data (e.g. social data such as customer reviews) be used to inform marketing campaigns and make improvements to the business?
- Can we use descriptive statistics to provide insights into the suitability of the loyalty points data to create predictive models?

Appendix B: Regression statistics

Spending vs Loyalty

Estimated parameters:

Intercept: -75.05266293364707

Coefficient: 33.06169325867294

Note: Standard errors not available in sklearn - use statsmodels for statistical significance

Predicted values (first 10):

[1214.35337415 2602.94449102 123.31749662 2470.69771798 1247.41506741
2437.63602473 123.31749662 3032.74650338 24.13241684 2305.38925169]

Coefficient: 33.06

R-squared (approx): 0.452

Income vs Loyalty

Estimated parameters:

Intercept: -65.68651279500432

Coefficient: 34.1878254856689

Note: Standard errors not available in sklearn - use statsmodels for statistical significance

Predicted values (first 10):

[354.82374068 354.82374068 382.85775758 382.85775758 410.89177448
410.89177448 438.92579137 438.92579137 466.95980827 466.95980827]

Coefficient: 34.19

R-squared (approx): 0.380

Age vs Loyalty

Estimated parameters:

Intercept: 1736.517739399063

Coefficient: -4.012805149995269

Note: Standard errors not available in sklearn - use statsmodels for statistical significance

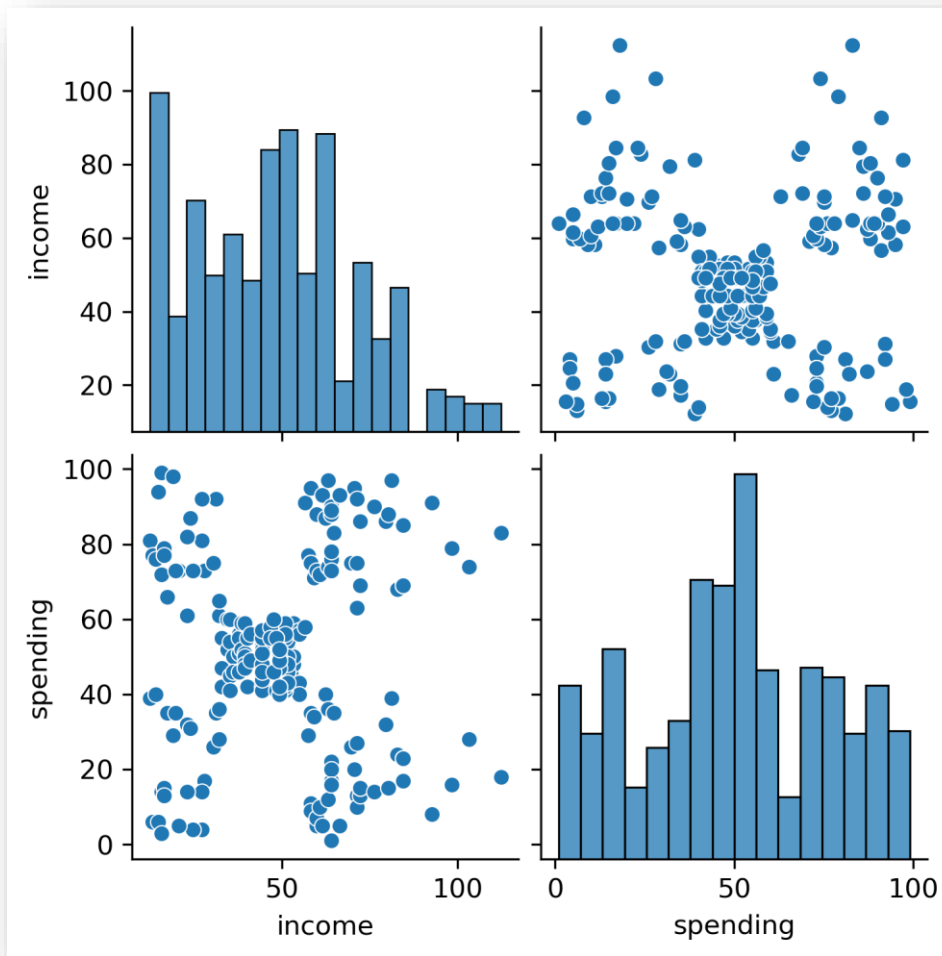
Predicted values (first 10):

[1664.2872467 1644.22322095 1648.2360261 1636.19761065 1604.09516945
1640.2104158 1588.04394885 1636.19761065 1471.6725995 1608.1079746]

Coefficient: -4.01

R-squared (approx): 0.002

Appendix C: More clustering data



		spending			income			loyalty	age		
cluster	count	mean	min	max	mean	min	max	mean	mean	min	max
0	356	82.01	63	97	73.24	56.58	112.34	3988.24	35.59	18	72
1	774	49.53	34	61	44.42	31.98	62.32	1420.38	42.13	17	72
2	330	17.42	1	39	74.83	57.4	112.34	911.76	40.67	17	72
3	269	79.42	61	99	20.35	12.3	31.98	971.94	31.6	17	71
4	271	19.76	3	40	20.42	12.3	31.98	275.06	43.51	17	72

Appendix D: Segmentation descriptions & assumptions

Develop the customer segments into distinct personas using results from the analysis and other available material.

Cluster_0: Executive Collectors; High Income, High Spend

TG is described as follows;

“The company manufactures and sells its own products, along with sourcing and selling products manufactured by other companies. Its product range includes books, board games, video games, and toys.”

Assume that this group is attract by specialist items that TG manufactures itself for the adult collector market as well as role play games and high end collectibles manufactured by suppliers like Showcase and Lego.

This is very much a specialist group of collectors and gamers with evidence in survey comments;

“Wrath of Ashardalon great investment for an avid RPG fan who enjoys tabletop as well as CRPGs”

Cluster_1: Gift Shoppers; Middle Income, Mid Spend

Busier individuals with slightly higher than average incomes buying toys as gifts for children in their extended family. They are more discerning and generous than groups spending less and will make more effort to buy specific items that appeal to the children they are buying for, including TG products for older teens.

“Purchased as a gift for my great nephew who is fascinated by equipment/trucks/tractors.”

Cluster_2: Selective Spenders; High Income, Low Spend

Every child’s least favourite aunt or uncle. They buy token gifts for young relatives with little thought or imagination in their selection. Likely to ignore or forget birthdays and buy only at Christmas. Buying 3rd lower quality third party children’s toys and games, not TG’s specialist collectibles.

Cluster_3: Generous Parents; Low Income, High Spend

Income likely to be linked to the early stage of their careers, coinciding with the raising of a young family. Generous gifters, they are likely to be very aware of their children's interests and desires and enjoy spending a higher proportion of their disposable income on their children. Susceptible to pester power but appreciative of good value and of good quality, well made products which can be handed down to younger siblings when the original recipient has outgrown them. Buying premium third party children's toys and games, not TG's specialist collectibles.

"Great quality very cute and perfect for my toddler"

"He loves construction vehicles so I am sure he will love this also"

Cluster_4: Budgeting Parents; Low Income, Low Spend

Unable or unwilling to spend larger amounts of their income on gifts for their children. They are likely to be less engaged with their children's interests and buy low quality items with little thought for how they will be received or their longevity as toys that can be played with repeatedly. Buying lower quality third party children's toys products, not TG's specialist collectibles.

"Loves stickers and she loves Peppa so this was perfect"

Appendix E: Decision tree text format

Cluster 0

```
|--- income <= 74.21
| |--- spending <= 84.50
| | |--- income <= 63.55
| | | |--- value: [2977.36]
| | |--- income > 63.55
| | | |--- value: [3215.75]
| |--- spending > 84.50
| | |--- income <= 68.47
| | | |--- value: [3742.98]
| | |--- income > 68.47
| | | |--- value: [4367.45]
|--- income > 74.21
| |--- spending <= 71.50
| | |--- income <= 83.64
| | | |--- value: [3716.89]
| | |--- income > 83.64
| | | |--- value: [3995.80]
| |--- spending > 71.50
| | |--- income <= 80.77
| | | |--- value: [4637.73]
| | |--- income > 80.77
| | | |--- value: [5288.98]
```

Cluster 3

```
|--- income <= 18.04
| |--- spending <= 96.50
| | |--- income <= 14.35
| | | |--- value: [597.55]
| | |--- income > 14.35
| | | |--- value: [750.35]
| |--- spending > 96.50
| | |--- value: [1027.36]
|--- income > 18.04
| |--- spending <= 74.00
| | |--- income <= 26.24
| | | |--- value: [968.17]
| | |--- income > 26.24
| | | |--- value: [1231.67]
| |--- spending > 74.00
| | |--- income <= 30.75
| | | |--- value: [1224.02]
| | |--- income > 30.75
| | | |--- value: [1637.00]
```